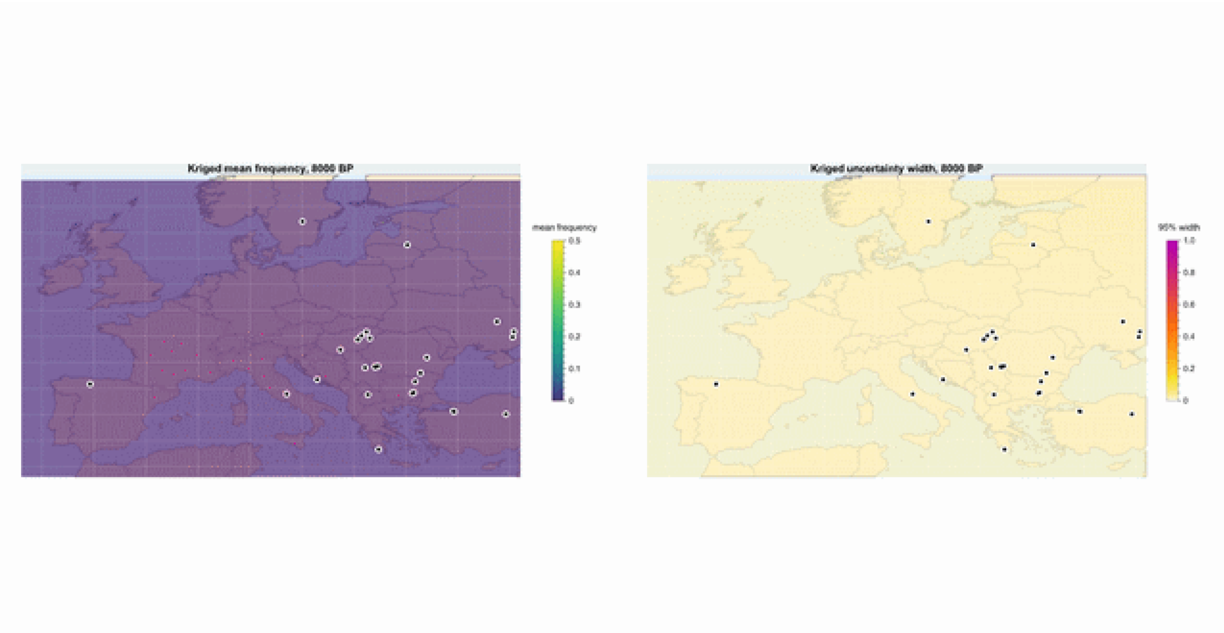
Mapping the rise of lactase persistence in ancient European DNA

A spatial diffusion–selection model fitted with sequential Monte Carlo ABC, in Wolfram Language

Wolfram Community submission — August 2026



Cover animation. Posterior mean rs4988235-derived allele frequency (left) and 95% credible-interval width (right) over Europe, 8000 BP to the present in 500-year steps. White-ringed dots are the ancient samples within ±500 years of each frame. Ordinary kriging is a display interpolation over the coarse 4° inference grid; the uncertainty panel is rendered at equal rank with the mean, so confidence can be read off wherever the eye lands.

Abstract

Lactase persistence is the textbook example of recent, strong selection in humans, and ancient DNA has turned its history into a quantitative time series: the derived allele tagged by `rs4988235` (the −13910*T variant near `LCT/MCM6`) is essentially absent in Mesolithic Europe, still rare in the Bronze Age, and reaches 50–90% in parts of north-western Europe today. This notebook builds an end-to-end Wolfram Language pipeline on the public GLAD ancient genotype workbook (2,999 samples, 1,785 with a usable call, age, and location): it reproduces the regional rise with binomial logistic fits, embeds it in an explicitly spatial

diffusion–selection model on a European grid with a dairying-onset covariate, and fits that model with SMC-ABC — 400 particles, adaptive tolerances falling 0.245 → 0.044 over five generations, 10,000 forward simulations. Posterior predictive coverage is 0.97; held-out-region and held-out-time validation and a five-scenario prior sensitivity analysis are reported alongside every point estimate. The posterior puts dairying-modulated selection at roughly 1–2% per generation (95% interval 0.001–0.036) with northern regional multipliers above one — and, echoing Evershed et al. (2022), it cannot cleanly separate milk-use-modulated from uniform selection.

1. Why lactase persistence

Two facts make this system scientifically interesting rather than merely illustrative. First, direct ancient-DNA time series show the allele was still rare long after dairying became widespread: **Evershed et al. (2022, Nature)** found that a selection model varying with prehistoric milk exploitation explains the allele trajectories *no better* than uniform selection since the Neolithic — famine and pathogen exposure, not everyday milk drinking, are their preferred drivers. Second, the implied selection is very strong: **Burger et al. (2020, Current Biology)** estimate a selection coefficient of about 0.06 for rs4988235-A over the last 3,000 years, starting from a Bronze Age frequency of only ~7%. A spatial model fitted honestly to the ancient genotypes should therefore (i) recover few-percent selection, (ii) show whether the data can tell dairying-modulated from uniform selection at all, and (iii) be loud about where its uncertainty lives.

2. The data

The input is the public GLAD ancient genotype workbook derived from the Allen Ancient DNA Resource v44.3 and used for the Evershed et al. (2022) analysis. The retrieval function downloads the workbook, records a SHA-256 checksum and a provenance manifest, and marks the raw file read-only. Everything below is reproducible from the repository with two wolframscript commands (§10).

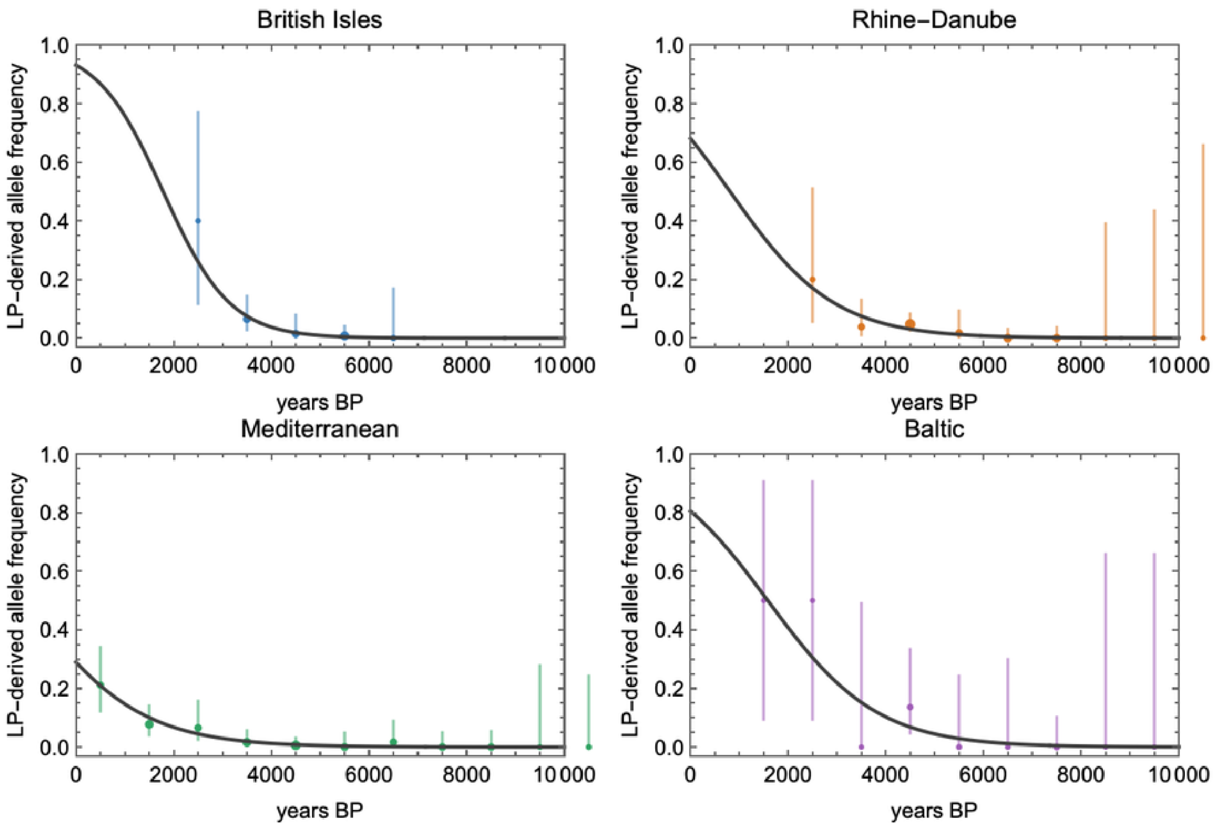
```
Get[FileNameJoin[{Directory[], "src", "LactasePersistenceSpatial.wl"}]];
raw = RetrieveRawData[Directory[]];
processed = WriteProcessedData[Directory[], raw];
samples = processed["CalledSamples"];
Length[samples]
```

Of 2,999 rows, **1,785 samples** carry a genotype call at rs4988235 together with usable age and coordinates. Genotype strings are normalized so that A (and the strand-flipped T, which occurs in 3 heterozygous calls) count as derived and G/C as ancestral; single-letter pseudo-haploid calls contribute one called allele, diploid calls two. Coordinates reported in millidegrees are rescaled, and each sample is assigned to one of four coarse analysis regions echoing the published framing: British Isles (171 samples / 304 called alleles / 9 derived), Rhine–Danube (341 / 591 / 16), Mediterranean (486 / 837 / 28), Baltic (56 / 90 / 5). Two properties of these data discipline everything that follows: derived alleles are rare every-where before ~4,000 BP, and sampling density collapses after ~2,500 BP, so any statement about the last two millennia is an extrapolation and is labelled as such.

3. Reproducing the regional trajectories

Before adding spatial structure, each region gets the standard non-spatial description: a binomial logistic trajectory `logit p(t) = α + β (10000 - t)/1000`, fitted by direct maximization of the binomial log-likelihood over called alleles (no time binning enters the fit; bins are display only). Standard errors come from the numerical Hessian at the optimum, and the fitter flags any solution on a parameter bound — an earlier iteration of this project silently reported a boundary solution for the British Isles, which is exactly the failure mode the flag now catches.

```
fits = FitAllRegionalLogistics[samples];
ExportRegionalFitOutputs[Directory[], samples, fits];
Dataset[fits]
```



Observed regional binned frequencies (points sized by called-allele count, 95% Wilson intervals) with fitted binomial logistic trajectories. Slopes per millennium (±1 SE): British Isles 1.46 ± 0.51, Rhine–Danube 0.94 ± 0.27, Mediterranean 0.86 ± 0.16, Baltic 0.90 ± 0.31. With a 28-year generation time these translate to approximate per-generation selection coefficients of 0.041, 0.026, 0.024, 0.025 — the familiar few-percent picture, strongest in the north-west, compatible with Burger et al.'s 0.06 for the last 3,000 years.

4. The spatial model

Europe (35–63°N, 12°W–35°E) is discretized into a 4° grid of 84 cells. Each cell carries a regional dairying-onset time (Mediterranean 8200 BP, Rhine–Danube 7600, British Isles

6100, Baltic 5600, other 6500) smoothed into a covariate `D(t) = 1/(1 + exp((t - onset)/350))`. Local derived-allele frequency evolves in 250-year steps from 10,000 BP to the present:

```
p[i, t+dt] = p[i, t] + g ( (s0 + sDairy D[i, t] m[region i]) p (1 - p)
                             + mig mean[ p[j, t] - p[i, t], j adjacent to i ] )

g = dt / 28   (generations per step)
```

with per-region selection multipliers `m` and a small latitudinal/longitudinal gradient allowed in the initial condition. One forward simulation costs ~26 ms, which is what makes honest simulation-based inference affordable: the full analysis uses ~40,000 simulations in under 20 minutes on a laptop. The model keeps demography implicit — it is a reaction–diffusion caricature whose job is to let the data say how much spatial structure, movement, and dairying-modulation they can actually constrain.

```
grid = BuildEuropeGrid[];
traj = SimulateSpatialTrajectory[
  <|"InitialFrequency" -> 0.003, "SelectionBase" -> 0.002,
    "SelectionDairying" -> 0.02, "Migration" -> 0.003|>, grid];
traj["TimesBP"] // Length
```

5. SMC-ABC inference

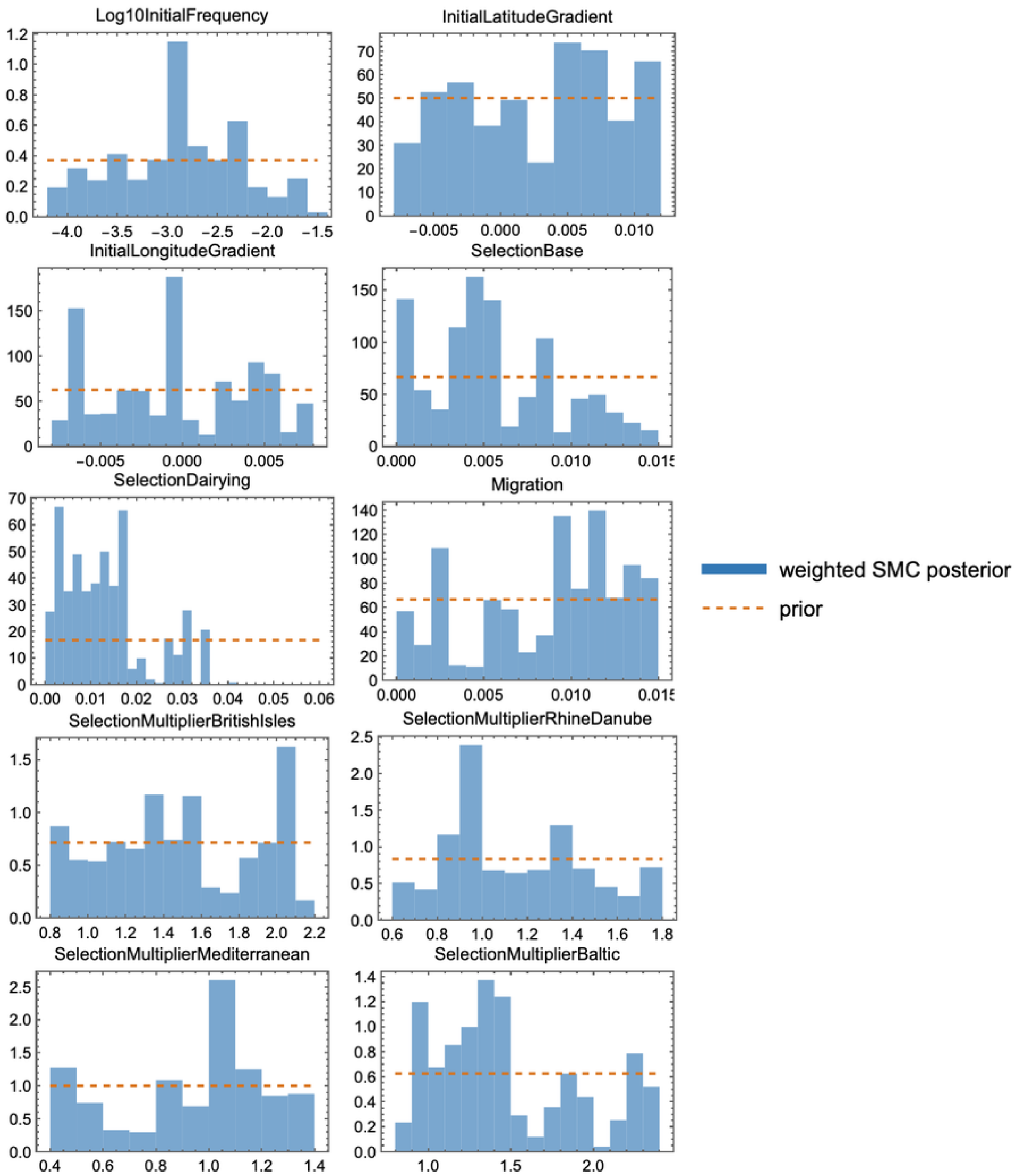
The likelihood is intractable in useful form, so the fit is simulation-based, comparing two families of summary statistics between data and simulation: (i) called-allele-weighted **regional time-binned frequencies** (39 bins), and (ii) **spatial gradients** — because a spatial model fitted only to regional aggregates cannot identify migration, each sample is linked to its nearest grid cell and time step, and pooled north–south and west–east frequency contrasts over the last 4,000 years are computed identically for observed alleles and model expectations.

Inference is sequential Monte Carlo ABC: 400 particles, 5 generations, tolerance set each generation to the median of the previous population's distances, Gaussian perturbation kernels with twice the weighted component variance, importance weights `w ∝ prior / Σ w K`, and ESS tracking. One implementation detail worth flagging for anyone building something similar: with a 10-dimensional box prior, perturbing in the original coordinates wastes ~97% of proposals outside the box (0.7^10 ≈ 0.03). The sampler therefore works in logit-transformed coordinates, where a uniform prior becomes an exact product of standard logistic densities, every proposal is automatically valid, and the prior density stays closed-form.

```
smc = RunSMCABC[samples, grid, "Particles" -> 400, "Generations" -> 5,
  "ProgressFunction" -> Function[{g, eps, acc, ess},
    Print["generation ", g, ": epsilon ", eps, ", ESS ", ess]]];
draws = ResamplePosterior[smc, 100];
ExportSMCOutputs[Directory[], samples, grid, smc, draws];
Dataset[PosteriorParameterQuantiles[smc]]
```

Priors are uniform: `log10 p0 ∈ [-4.2, -1.5]`, `s0 ∈ [0, 0.015]`, `sDairy ∈ [0, 0.06]` (spanning the published estimates), `mig ∈ [0, 0.015]`, regional multipliers within [0.4, 2.4]. The tolerance fell from 0.245 (prior median) to 0.044 over five generations and 10,000 simulations. The final effective sample size is 35 of 400 particles — importance-weight degeneracy is the known price of this weight formula when the posterior is much tighter than the prior, and it is reported rather than hidden; every interval below is a weighted quantile and should be read as approximate.

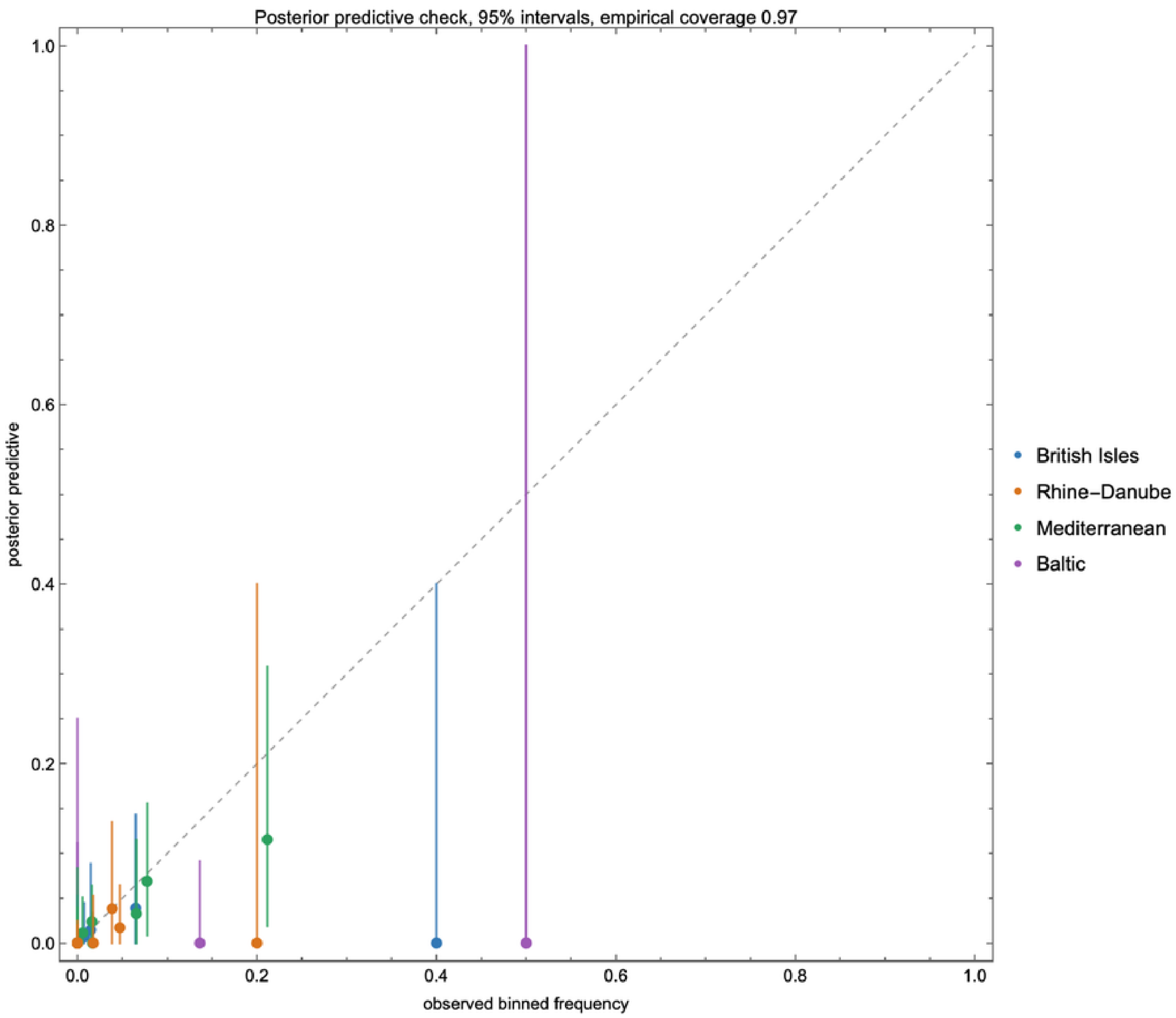
6. Results



Weighted SMC posterior (blue) against the flat prior (dashed orange) for all ten parameters. The informed dimensions are visible at a glance: initial frequency (posterior median p0 ≈ 0.0012), dairying-modulated selection sDairy (median 0.012, 95% interval [0.0014, 0.036]), baseline selection s0 (median 0.005 [0.0004, 0.013]), and the northern multipliers (British Isles 1.45, Baltic 1.37). The initial-gradient dimensions remain prior-dominated — equally visible, equally honest.

Three readings. First, total effective selection in a northern dairying cell at the posterior median, `s0 + sDairy × m ≈ 0.02` per generation, sits comfortably in the published few-percent range, with the northern multipliers pulled above 1 exactly where the logistic slopes were steepest. Second, the posterior does *not* cleanly separate dairying-modulated from baseline selection: the dairying covariate saturates within a few centuries of onset, so post-onset it acts almost like uniform selection — our small-model echo of Evershed et al.'s conclusion that milk-use-varying selection is not required by the allele trajectories. Third, migration is only weakly identified (§8).

7. Validation



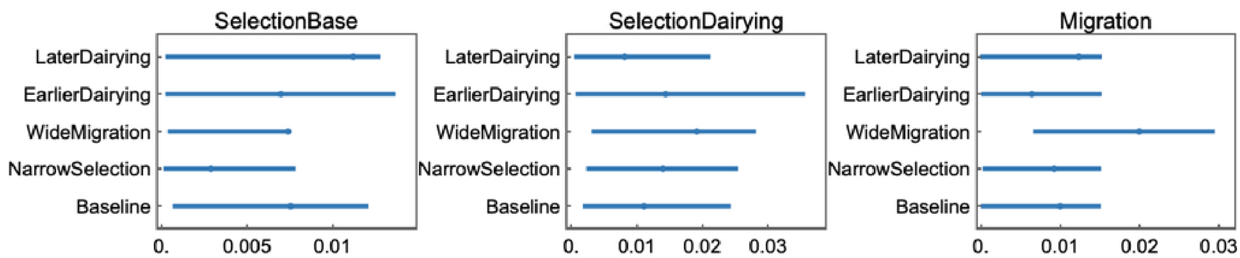
Posterior predictive check over the 37 regional time bins: posterior draws → forward simulation → binomial sampling at each bin's allele count. Empirical 95% coverage is 0.97 (36/37). Bins with tiny allele counts (e.g. Baltic bins with n = 2) correctly produce near-full-range intervals rather than sharp wrong predictions.

Held-out regions. Refitting with each region excluded and predicting its trajectory gives posterior-median RMSEs of 0.047 (Mediterranean), 0.065 (Rhine–Danube), 0.173 (Baltic), 0.180 (British Isles), with the 95% predictive intervals covering all held-out bins — wide where data are thin, rather than sharply wrong.

Held-out time. The most severe test: train only on samples older than 2,500 BP (1,543 samples) and predict the 7 most recent bins. The intervals still cover, but the median under-predicts the recent rise badly (RMSE 0.32). This is not a bug — it is the pipeline rediscovering, from deep-past data alone, that most of the evidence for strong late selection lives in the last three millennia, precisely the period Burger et al. (2020) analysed on independent grounds.

```
cv = RunSMCCrossValidation[samples, grid];
timeSlice = RunTimeSliceValidation[samples, grid];
Dataset[cv]
```

8. Sensitivity



Posterior medians and 95% intervals under five scenarios: baseline, halved selection priors, doubled migration prior, and dairying onsets shifted ±400 years. The dairying-modulated selection component stays positive throughout (medians 0.008–0.019). The migration posterior tracks its prior (median 0.010 baseline → 0.020 under the widened prior): with 84 cells and aggregated summaries, movement is weakly identified, and the pipeline says so rather than dressing a prior up as a finding.

9. What this is not

The regional logistic layer is a qualitative reproduction of the published four-region picture, not a parameter-for-parameter replication of Evershed et al.'s analysis. The spatial model is a coarse caricature with implicit demography and a deterministic core; the final ESS is modest; and everything after ~2,000 BP extrapolates beyond the ancient data. Obvious next steps: a stochastic (Wright–Fisher noise) simulator, ancestry-aware likelihoods at the sample level, an SMC sampler with MCMC rejuvenation moves to fight weight degeneracy, and radiocarbon-calibrated date uncertainty propagated into the bins.

10. Reproducibility

Everything — retrieval, processing, fits, SMC-ABC, validation, sensitivity, maps, animation, and this notebook — is generated by the repository at **mthiel74/ancient-dna-lactase-spatial-wolfram** (`https://github.com/mthiel74/ancient-dna-lactase-spatial-wolfram`):

```
wolframscript -file scripts/retrieve_data.wls
wolframscript -file scripts/run_pipeline.wls --particles 400 --generations 5
wolframscript -file scripts/run_tests.wls
wolframscript -file community/build_notebook.wls
```

The package `src/LactasePersistenceSpatial.wl` (~1,700 lines) exposes the full workflow; `notebooks/LactasePersistenceSpatial.nb` walks the same pipeline interactively; 24 VerificationTests cover parsing, the simulator, the summary statistics, and the SMC machinery, and GitHub Actions runs them non-interactively. Raw data are preserved read-only with checksums; every processed table carries provenance.

11. References

Evershed, R. P., et al. (2022). Dairying, diseases and the evolution of lactase persistence in Europe. *Nature* 608, 336–345. `https://www.nature.com/articles/s41586-022-05010-7`

Burger, J., et al. (2020). Low prevalence of lactase persistence in Bronze Age Europe indicates ongoing strong selection over the last 3,000 years. *Current Biology* 30(21), 4307–4315. `https://www.cell.com/current-biology/fulltext/S0960-9822(20)31187-8`

GLAD LP Ancient Genotypes 2022 workbook (UCL), derived from the Allen Ancient DNA Resource v44.3. `https://www.ucl.ac.uk/biosciences/sites/biosciences/files/glad_adna_15-8-22.xlsx`

Generated on August 31, 2026.